Flavor Perception AI

Digitizing the Unseen Senses

Research Interest Presentation // MIT 2026

COFFEE SHOP ROOTS

During my gap year at MIT, I worked as a barista. It wasn't just about the caffeine; it was a deep dive into the **complexity of flavor**.

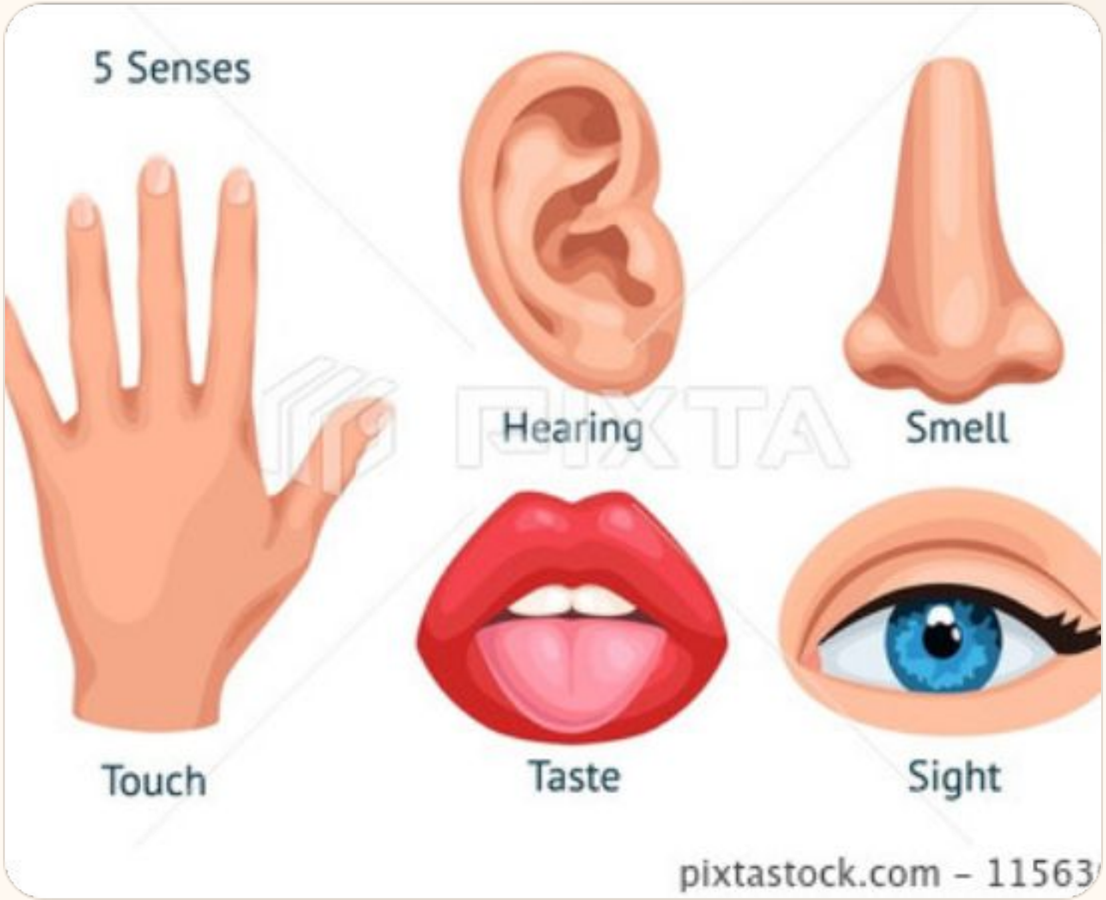
I noticed how temperature, acidity, and chemical extraction changed every cup. It sparked a question: *How do we translate this biological experience into data?*

THE SENSORY GAP IN AI

AI has mastered the "Digital Senses":

- **Vision:** Object detection & recognition
- **Audio:** NLP and voice synthesis
- **Touch:** Haptic feedback robotics

But **Smell** and **Taste** remain largely analog and subjective.



DIGITIZING THE "CODE OF SMELL"

The Language Problem

Smell is currently described with subjective words like "floral" or "woody." These vary by individual and culture.

The Biological Code

Human noses use ~400 receptors. Perception is actually a combinatorial "barcode" of molecular triggers.

The Digital Standard

We are building a "Pantone for Scent"—a mathematical representation that AI can read and predict.

CHATGPT FOR CHEMICALS

Instead of training models on English, we train them on **Chemical Data**.

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Molecules = Words
 Chemical Bonds = Grammar
 Flavor Profile = Narrative
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By mapping molecular structures to sensory outcomes, AI can "hallucinate" entirely new flavors or aromas.

THE CHEMICAL VOCABULARY

40B

Predicted Odorant Molecules

100M

Molecules Known to Science

Mapping the Uncharted

The chemical space is vast. Our foundation model bridges the gap between:

- **Molecular Parameters:** Boiling point, functional groups.
- **Sensory Descriptors:** Intensity, sweetness, and emotional response.

FOOD & BEVERAGE INNOVATION

Rapid R&D: Compressing flavor development cycles from years to months.

Ingredient Substitution: Finding "sensory-equivalent" replacements for scarce natural extracts.

Masking: Blocking bitter aftertastes in plant-based proteins using molecular antagonists.

FRAGRANCE & PERSONAL CARE

Programmable Scent

Startups are now using receptor-level modeling to engineer custom scent molecules.

This enables **sustainable alternatives** to wild-harvested botanicals and precision customization for individual consumer preferences.

HEALTH & NUTRITION

Diet Adherence

Personalized plans that match nutritional needs with individual flavor preferences.

Sugar Reduction

Using AI to "hack" perception—enhancing sweetness without adding actual glucose.

Behavioral Alignment

Moving to continuous engagement through sensory-aligned recommendations.

BROAD SCIENTIFIC HORIZONS

The chemical foundation model applies far beyond the kitchen:

- **Pharmaceuticals:** Predicting molecular docking and masking drug bitterness.
- **Environment:** Detecting pollutants or identifying disease "scents" in the air.
- **Biotech:** Engineering molecules to repel disease-carrying insects.

MAPPING SENSES TO AGI

General intelligence is fundamentally the process of **mapping sensory inputs to behavior**.

To achieve truly embodied Artificial General Intelligence (AGI), machines must perceive the world with the same **five-sense fidelity** as biological organisms.

Understanding taste and smell is not optional—it is necessary to decode the full chemical reality of our environment.

Questions?

Let's discuss the future of sensory intelligence.

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IMAGE SOURCES

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